NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Course Name: ETHICAL HACKING

Assignment- Week 1

TYPE OF QUESTION: MCQ/MSQ/SA

Number of questions: 10 **Total marks: 10 x 1 = 10**

QUESTION 1:

Which of the following belong to the CIA Triad in information security?

a. Confidentiality
b. Authentication
c. Availability
d. Accountability
e. Integrity
f. Authorization

Correct Answer: a, c, e

Detail Solution: The CIA Triad consists of Confidentiality, Integrity, and Availability. Authentication, non-repudiation, accountability, and authorization are important security concepts but are not part of the core CIA Triad.
Thus, the correct options are (a), (c), and (e).

QUESTION 2:

Which of the following penetration testing models provides the tester with no prior information about the target network?

a. White box model
b. Black box model
c. Gray box model
d. Blue box model

Correct Answer: b

Detail Solution: In the white box model, the tester has complete information about the network. In the black box model, the tester has no information about the network. The grey box model provides partial information. There is no standard model called the blue box model. Thus, the correct option is (b).

QUESTION 3:

Which of the following statement(s) is/are true for packet-switched networks?

a. Bandwidth is dedicated for the entire duration of a session.
b. Multiple connections can share the same communication link.
c. Delay is always constant regardless of network load.
d. It requires end-to-end circuit setup before data transfer.
e. Data is divided into smaller packets before transmission.
f. It guarantees zero packet loss under all conditions.

Correct Answer: b, e

Detail Solution: In packet switching, communication links may be shared by more than one connection, and data is broken into smaller packets before transmission. Dedicated bandwidth and constant delay are characteristics of circuit switching. Packet switching does not require prior circuit setup, and it does not guarantee zero packet loss.
Thus, the correct options are (b) and (e).

QUESTION 4:

Which OSI layer is primarily responsible for framing, MAC addressing, and error detection over a physical link?

a. Physical layer
b. Data link layer
c. Network layer
d. Transport layer
e. Session layer

Correct Answer: b

Detail Solution: The data link layer organizes bits into frames, provides MAC addressing, and performs error detection over a point-to-point or shared link. The physical layer deals with raw bit transmission, and the network layer handles routing and logical addressing.

Thus, the correct option is (b).

QUESTION 5:

A 1500-byte packet is sent over a 100 Kbps point-to-point link whose propagation delay is 8 msec. The packet will reach the destination after _____ msec. (Assume 1K = 1000)

Correct Answer: 120 to 130

Detail Solution: Transmission rate = 100,000 bits per second.
1500 bytes = 12,000 bits.
Transmission time = 12,000 / 100,000 = 0.12 sec = 120 msec.
Total time = 120 + 8 = 128 msec.

QUESTION 6:

What is the primary purpose of an IP address in a network?

a. It identifies the physical hardware manufacturer of a device.
b. It uniquely identifies a host or interface on a network.
c. It specifies the number of open ports on a system.
d. It determines the encryption algorithm used by a host.

Correct Answer: b

Detail Solution: An IP address is a logical address that uniquely identifies a host or a specific network interface, enabling routing of packets to the correct destination. It is unrelated to hardware manufacturer identification, open ports, or encryption.
Thus, the correct option is (b).

QUESTION 7:

_____ is the minimum size (in bytes) of a TCP header without options.

Correct Answer: 20

Detail Solution: The minimum size of a TCP header is 20 bytes when no options are used. Additional options can increase this size up to 60 bytes.

Thus, the correct answer is 20.

QUESTION 8:

Which IP header field is decremented at every router hop and is primarily used to prevent packets from looping indefinitely in a network?

a. Header checksum
b. Fragment offset
c. Time to Live (TTL)
d. HLEN
e. Identification

Correct Answer: c

Detail Solution: The Time to Live (TTL) field is decremented by 1 at every router hop. When it reaches 0, the packet is discarded, preventing it from looping indefinitely in the network. Header checksum is used for error detection, fragment offset and identification are used for reassembly, and HLEN indicates header length.

Thus, the correct option is (c).

QUESTION 9:

If the IP header is 224 bits long, what will be the value (in decimal) of the HLEN field?

Correct Answer: 7

Detail Solution: HLEN represents the header length in multiples of 32 bits (4 bytes).

Thus, 224 bits = 7 x 32 bits. Hence HLEN = 7.

Thus the correct answer is 7.

QUESTION 10:

_____bits are used to represent a MAC address.

Correct Answer: 48

Detail Solution: A MAC address (data link layer / hardware address) is 48 bits long, typically represented as 12 hexadecimal digits.
Thus, the correct answer is 48.

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